Why impact also needs energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why momentum does not tell the whole impact story.

SUCCESS CRITERIA - I can...

1. Compare momentum and kinetic energy as different quantities.
2. Identify speed as a squared factor in kinetic energy.
3. Name where energy can go in an impact.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “kinetic energy” to explain your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

kinetic energy: energy associated with motion

scalar: a quantity described by size without a vector direction

01 Warm-up: bring back an earlier idea

A and B have equal momentum magnitude, but A has 3 times B’s mass.
Compare speeds and kinetic energies algebraically.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p = mv$; $K = \frac{1}{2}mv^2$.
Momentum includes direction; kinetic energy is a scalar. Neither alone gives a complete damage forecast.

Worked reasoning: A 2 kg cart at 3 m/s has momentum 6 kg·m/s and kinetic energy $\frac{1}{2}\times2\times3^2=9$ J. A 1 kg cart at 6 m/s has the same momentum but 18 J of kinetic energy.

Your turn: Draw labeled momentum arrows and energy bars for the two carts.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two equal-mass carts move at speeds 2 and 4 m/s.
Predict how their momenta and kinetic energies compare before calculating.

Sentence starter: Both … . They differ in … because … .



Same mass; 2 m/s

Same mass; 4 m/s

Marzano check - circle: 1 2 3

04 Apply the physics

Two lab carts each have momentum magnitude 12 kg·m/s. Their masses are 2 kg and 6 kg.
Find each speed and kinetic energy.
Draw energy bars and explain why equal momentum does not guarantee equal kinetic energy.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Under the assumption …, the evidence … supports … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

List three forms or destinations of energy after a deforming collision. Does “less kinetic energy” mean “energy destroyed”?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A positive triangular force pulse lasts 0.4 s and peaks at 60 N.
Find impulse and velocity change for a 0.8 kg object.
Show the signed impulse on the force-time model.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found … [unit]. This value means … in this situation.

Force (N)



Time (s)

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

At fixed momentum, mass triples. Derive or explain the kinetic-energy multiplier using $K=p^2/(2\ m)$.
Use the energy model to show the quantities you compare; label units and any unknowns.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two carts have equal momentum magnitude p. Derive K in terms of p and m, then compare energies if one mass is four times the other.
Use the energy model to show the quantities you compare; label units and any unknowns.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3